

The Stream Cipher HC-128

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Abstract. We present the 128-bit version of the stream cipher HC-256.

1 Introduction

The stream cipher HC-128 is a simplified version of HC-256 [15] for 128-bit security. HC-128 is a simple, secure, software-efficient cipher¹ and it is freely-available. HC-128 consists of two secret tables, each one with 512 32-bit elements. At each step we update one element of a table with non-linear feedback function. All the elements of the two tables get updated every 1024 steps. At each step, one 32-bit output is generated from the non-linear output filtering function. HC-128 is suitable for the modern (and future) super-scalar microprocessors. The dependency between operations in HC-128 is very small: three consecutive steps can be computed in parallel; at each step, the feedback and output functions can be computed in parallel. The high degree of parallelism allows HC-128 to run efficiently on modern processors. We implemented HC-128 in C, and the encryption speed of HC-128 reaches 3.05 cycles/byte on the Intel Pentium M processors.

HC-128 is very secure. Our analysis shows that recovering the key of HC-128 is as difficult as exhaustive key search. To distinguish the keystream from random, we expect that more than 2^{64} keystream bits are required (our current analysis shows that about 2^{151} outputs are needed in the distinguishing attack). This report is organized as follows. We introduce HC-128 in Section 2. The security analysis of HC-128 is given in Section 3 and Section 4. Section 5 discusses the implementation and performance of HC-128. Section 6 concludes this report.

2 Cipher Specifications

In this section, we describe the stream cipher HC-128. From a 128-bit key and a 128-bit initialization vector, it generates keystream with length up to 2^{64} bits.

¹ This description of the cipher was prepared by the editor using the version available at <http://www.ecrypt.eu.org/stream/> as a source. Any transcription errors are the fault of the editor and the reader is recommended to consult the original.

2.1 Operations, Variables, and Functions

The following operations are used in HC-128:

- $+$ $x + y$ means $x + y \bmod 2^{32}$, where $0 \leq x < 2^{32}$ and $0 \leq y < 2^{32}$
- $\boxminus$ $x \boxminus y$ means $x - y \bmod 512$
- $\oplus$ bit-wise exclusive OR
- $\parallel$ concatenation
- $\gg$ right shift operator: $x \gg n$ means x being right shifted n bits
- $\ll$ left shift operator: $x \ll n$ means x being left shifted n bits
- $\ggg$ right rotation operator: $x \ggg n$ means $((x \gg n) \oplus (x \ll (32 - n)))$, where $0 \leq n < 32$, $0 \leq x < 2^{32}$
- $\lll$ left rotation operator: $x \lll n$ means $((x \ll n) \oplus (x \gg (32 - n)))$, where $0 \leq n < 32$, $0 \leq x < 2^{32}$

Two tables P and Q are used in HC-128. The key and the initialization vector of HC-128 are denoted as K and IV. We denote the keystream being generated as s .

- P a table with 512 elements of 32-bits denoted by $P[i]$ for $0 \leq i \leq 511$
- Q a table with 512 elements of 32-bits denoted by $Q[i]$ for $0 \leq i \leq 511$
- K the 128-bit key of HC-128
- IV the 128-bit initialization vector of HC-128
- s the keystream being generated from HC-128; The 32-bit output of the i th step is denoted as s_i . Then $s = s_0 \parallel s_1 \parallel s_2 \parallel \dots$

There are six functions being used in HC-256. $f_1(x)$ and $f_2(x)$ are the same as the $\sigma_0^{256}(x)$ and $\sigma_1^{256}(x)$ being used in the message schedule of SHA-256 [14]. For $h_1(x)$, the table Q is used as S-box. For $h_2(x)$, the table P is used as S-box.

$$\begin{aligned}
 f_1(x) &= (x \ggg 7) \oplus (x \ggg 18) \oplus (x \gg 3) , \\
 f_2(x) &= (x \ggg 17) \oplus (x \ggg 19) \oplus (x \gg 10) , \\
 g_1(x, y, z) &= ((x \ggg 10) \oplus (z \ggg 23)) + (y \ggg 8) , \\
 g_2(x, y, z) &= ((x \lll 10) \oplus (z \lll 23)) + (y \lll 8) , \\
 h_1(x) &= Q[x_0] + Q[256 + x_2] , \\
 h_2(x) &= P[x_0] + P[256 + x_2] ,
 \end{aligned}$$

where $x = x_3 \parallel x_2 \parallel x_1 \parallel x_0$, x is a 32-bit word, x_0, x_1, x_2 , and x_3 are four bytes. The bytes x_3 and x_0 respectively denote the most and least significant byte of x .

2.2 Initialization Process (Key and IV Setup)

The initialization process of HC-128 consists of expanding the key and initialization vector into P and Q (similar to the message setup in SHA-256) and running the cipher 1024 steps (with the outputs being used to update P and Q).